

Optimal Time-Stepping with Non-Iterative Defect Correction

for the Moment Hierarchy of Stochastic Reaction–Diffusion Equations

A living paper: derivation, verification ledger, and an honest map from closed moment systems to Cole–Hopf and KPZ observables

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Repository: [OTS–NIDC](#) (this paper is generated from it; see Appendix [A](#))

Abstract

This paper extends the optimal time-step selection with non-iterative defect correction (OTS–NIDC) framework of Chu (2011) from deterministic reaction–diffusion equations to the *moment hierarchy* of stochastic reaction–diffusion equations with linear multiplicative noise. Because Itô’s formula closes the hierarchy exactly for affine drift with linear multiplicative noise, every moment equation is a deterministic reaction–diffusion PDE to which OTS–NIDC applies verbatim: a single optimal step $\Delta t_{\text{opt}} = \Delta x^2/(6D)$, independent of both the moment order and the noise amplitude, cancels the leading truncation term, and an explicit correction restores fourth-order global accuracy on a three-point stencil. We derive the scalar family, the coupled mean-field/second-moment systems, and the competing-time-step problem that arises between them, with resolution via decoupled blocks and multirate subcycling with an exact mismatch correction. All claims are entered in a *status ledger*: verified numerically (V), proven in sketch (P), stable-region supported (S), open (O), or contradicted by our own verification harness (C) — most notably, the coupled system currently measures at second order, not the asserted fourth, and we report this rather than bury it. The final section maps the road out of the linear world: Cole–Hopf makes the stochastic Burgers equation a multiplicative-noise stochastic heat equation for ϕ , so weakly convergent schemes for ϕ moments transfer to exponential functionals of Burgers — the log-normal observables that fail Gaussian closure — while the KPZ regime $u = \log \phi$ retains well-defined roughness observables in the same window where the ϕ -moments themselves diverge. We frame what is proved, what is plausible, and what must be rebuilt, and invite correction.

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How to read this paper (the living-paper contract)

This is a *living paper*: a submission-track manuscript that carries its own epistemic state. It will be re-released as derivations firm up, verification harnesses expand, and open items resolve. Two mechanisms enforce that:

1. **Status ledger.** Every substantive mathematical or numerical claim in the text carries one of five tags, with the tag defined by a reproducible artifact in the repository, not by confidence:

Tag	Meaning & evidence required
[V]	Verified. A convergence/stability number produced by a script in <code>code/</code> against an independent exa
[P]	Proven (sketch). A complete argument the reader can check line by line; no numerical artifact, but
[S]	Supported. Backed by an analysis artifact whose conclusion is weaker than the claim (e.g. scalar vo
[O]	Open. Stated as a target or conjecture; no artifact yet.
[C]	Contradicted. Our own harness produces numbers inconsistent with the claim. Kept in the text, w

2. **Change log.** Every release records what moved between states.

Status ledger (current release)

Claim (where)	Status	Artifact
Scalar OTS: $\Delta t_{\text{opt}} = \Delta x^2/(6D)$ cancels the u_{xxxx} LTE term (Thm. 1)	[P]	§3 derivation; <code>sign-change.tex</code>
Δt_{opt} independent of moment parameter α and noise amplitude σ (Cor. 3)	[P]	binomial identity, §3
$p = 1$ NIDC sign: $C^{(1)}$ must be <i>added</i> (Rem. 4)	[V]	<code>code/implementation/results.md</code> : 4.0
Scalar $p = 1$ global order 4 at Δt_{opt} (Thm. 5)	[V]	<code>results.md</code> ; 5-point-check harness
4th-order FD + plain FE caps at order 2 (same step)	[V]	<code>results.md</code>
Von Neumann: NIDC shrinks CFL by factor $1 - \eta/18 + O(\eta^2)$; stable at Δt_{opt}	[V]	<code>code/stability/stability_analysis.m</code>
Moment hierarchy closes for affine drift + linear multiplicative noise (Prop. 13)	[P]	§5; <code>moment_hierarchy_primer.tex</code>
M-block/V-block exact decoupling (Prop. 8)	[P]	<code>competing-timesteps-multirate.md</code> §2
2:1 step ratio in the moment hierarchy $D_i = D$	[V]	<code>notes/Di_eq_D_verification.md</code> : ratio
Regime boundary $r^* = 3$ from FTCS at shared step (Thm. 10)	[S]	derived; scalar FTCS argument (coupled
Regime 1 exact mismatch correction at any ϵ (Thm. 9)	[V]	<code>examples/test_regime1_D1.3.D2.1.py</code>
Remainder-step error uniformly $O(\Delta x^4)$ in $\text{frac}(r) \in [0, 1)$ (Thm. 11)	[P]	<code>competing-timesteps</code> §5.2 (leading term
Coupled M+V, separate stepping: 4th order in both (Cor. 12)	[C]	<code>coupled_verify.py</code> : measured order ≈ 2
$p \geq 2$ NIDC tower at $(2p + 3)$ -point stencil	[O]	stencil-bound argument in §3; no harness
Cole–Hopf transfer of weak schemes to Burgers functionals (Thm. 14)	[P]	§6; mild-solution representation
KPZ-regime divergence of raw ϕ -moments while u -observables stay finite	[S]	literature argument; no numerics in this
MLMC order- ≥ 2 weak base eliminates $\log \epsilon$ factor	[O]	cost argument, §7

Change log

v0.3 (2026-09-16). First public living-paper release. Full derivation arc assembled (scalar $\rightarrow$ coupled $\rightarrow$ stochastic $\rightarrow$ Cole–Hopf/KPZ outlook). Evidence audit performed: the 2:1 ratio claim independently re-verified; the coupled fourth-order claim *demoted to* [C] after `coupled_verify.py` measured order ≈ 2 ; the scalar sign correction (2026-07-09) incorporated throughout; stability direction corrected (NIDC shrinks, not expands, the CFL window).

v0.1–0.2 (2026-07). Internal working notes: derivation documents OTS-NIDC.md, multirate note, sign-change proof, stability analysis; scalar $p = 1$ implementation verified at 4th order; coupled harness built.

1 Introduction

1.1 The problem class

Consider the stochastic heat equation with linear multiplicative noise, in Itô form, on a one-dimensional domain with (say) homogeneous Neumann conditions:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + \sigma u \dot{W}(x, t), \quad \dot{W} = \text{space-time white noise.} \quad (1)$$

The field $u(t, x, \omega)$ is random at every point. The standard numerical route — Euler–Maruyama or Milstein on a spatial discretization, driven by sampled Wiener increments, followed by Monte Carlo averaging over many paths — is a pathwise method whose *purpose* is to estimate statistics like $m(x, t) = \mathbb{E}[u]$ and $V(x, t) = \mathbb{E}[u^2]$. It pays for those statistics M times over, once per path, and its weak order in time is at most 1 (Lord et al., 2014; Kloeden and Platen, 1992).

This paper pursues the complementary route: *do not discretize the noise at all*. For the model class of §5, Itô’s formula converts (1) into a closed, deterministic, infinite system of reaction–diffusion PDEs for the moments $M_k = \mathbb{E}[u^k]$. The moment equations are the deterministic object the Monte Carlo estimate was approximating; we solve them directly, at high order, with a single-pass explicit finite-difference scheme.

1.2 The method in one paragraph

The deterministic core is Chu’s OTS-NIDC (Chu, 2011): for a finite-difference time-stepping scheme, *choose the time step so that the leading truncation term cancels*, then add a non-iterative defect correction to cancel the surviving residual. For the reaction–diffusion family $u_t = Du_{xx} + \alpha u$ (constant coefficients), the optimal step is

$$\Delta t_{\text{opt}} = \frac{\Delta x^2}{6D}, \quad (2)$$

one third of the explicit CFL limit, and the first-level correction restores fourth-order global accuracy on the standard three-point stencil (Theorem 5). The stochastic content of this paper is the observation that *every* equation in the moment hierarchy of (1) is a member of this family, with $\alpha = \frac{k(k-1)}{2}\sigma^2$ for the k -th moment — so one optimal step serves all moments simultaneously, and the Itô correction terms enter the defect correction as explicitly computable coefficients (Corollary 3).

1.3 Why this is not too good to be true: scope, in both directions

Two boundary conditions on the claim must be stated up front.

Strong-convergence barriers do not apply. Davie and Gaines (Davie and Gaines, 2001) showed that any scheme that approximates the solution of a parabolic SPDE driven by space-time white noise *through sampled Wiener increments* cannot beat strong order 1/2 in time (for the natural norms). OTS-NIDC never samples the Wiener process: it solves deterministic PDEs

derived from the SPDE by Itô calculus. The barrier governs the pathwise problem, a different mathematical object. What the method buys is *weak-type accuracy in the moments* — fourth-order convergence of the computed moment fields to the true moment fields of the closed system — at the cost of abandoning pathwise information entirely.

Closure validity is a separate, upstream question. Fourth-order accuracy of the solver is a statement about the *closed* moment system. It converts to a statement about the true SPDE law only when the hierarchy actually closes — which, per Proposition 13, happens exactly for affine drift with additive or linear-multiplicative noise, and fails for polynomial reactions of degree ≥ 2 (Fisher–KPP) and for nonlinear noise (chemical Langevin). Grid refinement does not rescue a bad closure. Every stochastic application in this paper therefore lives inside the closure-exact class, and the nonlinear frontier (§6) reaches *beyond* it only through exact transforms (Cole–Hopf), not through moment truncation.

1.4 Contributions and structure

1. **Unified scalar theory** (§3): the p -level OTS-NIDC family for $u_t = Du_{xx} + \alpha u$, with the α -independence of Δt_{opt} (Prop. 2), the sign-corrected first-level correction (Remark 4, verified at 4.01 measured order), the stencil-scaling bound, and the von Neumann stability analysis showing the correction is stability-neutral to $O(\Delta x^2)$.
2. **Coupled deterministic reaction–diffusion** (§4): the mean-field (M) and second-moment (V) systems, exact decoupling of the two blocks, the competing-time-step theorem ($\Delta t_{\text{opt}}^{(ij)} = \Delta x^2/6(D_i + D_j)$), the regime boundary $r^* = 3$, the exact mismatch correction for shared stepping, and the subcycling/remainder-step construction with its uniform bound — plus the verification contradiction ([C]) that currently caps the coupled claim.
3. **Stochastic reaction–diffusion** (§5): the Itô moment hierarchy, exact closure, the identification $\alpha_k = k(k-1)/2 \cdot \sigma^2$, the two-block moment system for coupled stochastic species, and the interpretation of the method as a weak (moment) solver that side-steps the Davie–Gaines barrier by construction.
4. **The nonlinear frontier: Cole–Hopf and KPZ** (§6): what survives outside the closure-exact class. The Cole–Hopf transform maps stochastic Burgers to a multiplicative-noise stochastic heat equation, so our schemes for ϕ -moments transfer to exponential functionals of Burgers; the KPZ side retains well-defined roughness observables precisely in the window where raw ϕ -moments diverge — and we are explicit that this window is exactly where two-point closure is lost and the framework must be rebuilt.
5. **An honest ledger** (front matter, §7): every claim tagged, every state change logged, and the one [C]item — the coupled fourth-order assertion — displayed rather than resolved.

2 Relation to prior work

2.1 The OTS-NIDC lineage

The method descends from Chu (Chu, 2011), who combined optimal time-step selection with non-iterative defect correction to boost finite-difference schemes for linear and semilinear PDEs to fourth order on three-point stencils, with extensions to more dimensions and irregular domains. Chu and

Lambers (Chu and Lambers, 2009) extended OTS to spatially *variable* leading coefficients via an optimal-grid transformation, with the constant-coefficient step $\Delta t_{\text{opt}} = \Delta x^2/6D$ recovered when the coefficient is constant. Razi, Attar and Vedula (Razi et al., 2015) coupled grid adaptation with non-iterative defect correction; our critique notes (literature/Razi2015_critique.md) observe that their joint optimization degenerates to the OTS step for linear coupling operators — a point we return to when justifying why the coupled corrections of §4 need no further optimization.

The present work is, to our knowledge, the first systematic extension of this lineage to *stochastic* equations — accomplished not by discretizing a random field but by moving up one level of abstraction: the moment hierarchy of the SPDE is itself a family of deterministic reaction–diffusion equations, so the entire deterministic theory applies unmodified (§5).

2.2 SPDE numerics: the pathwise mainstream

The numerical SPDE literature is dominated by pathwise schemes. The textbook frame is Lord, Powell and Shardlow (Lord et al., 2014), building on Kloeden–Platen (Kloeden and Platen, 1992). Within it:

- **Euler–Maruyama and Milstein.** The canonical schemes, weak order 1 and strong order 1/2 (Milstein: strong order 1 for commutative noise) (Kloeden and Platen, 1992; Leonhard and Rößler, 2014; Haji-Ali and Stein, 2023). For SPDEs the spatial semi-discretization error typically dominates; strong rates are further limited by the spatial regularity of the noise. Every pathwise method pays a Monte Carlo factor $O(M^{-1/2})$ to reach moments. That is the competitor baseline for this paper: identical statistics, obtained M times over.
- **The Davie–Gaines barrier.** Davie and Gaines (Davie and Gaines, 2001) proved that numerical schemes for parabolic SPDEs driven by space-time white noise, converging through Wiener increments, cannot exceed strong order 1/2 in time. This is the fundamental obstruction to high-order *pathwise* schemes for (1) — and it is precisely what the moment-level reformulation sidesteps, by never entering the pathwise problem.
- **Weak-rate separation.** Bolin, Kirchner and Kovács (Bolin et al., 2018) and Cui–Sun (Cui and Sun, 2023) exemplify a robust phenomenon: weak rates can exceed strong rates for SPDE approximations. Our contribution pushes this to its structural limit: at the moment level, the problem is deterministic, so the “weak rate” is simply the accuracy of a deterministic solver — arbitrarily high, at the price of the closure.
- **Structure-preserving alternatives.** Exponential integrators treat the stiff linear part exactly (Lord and Tambue, 2010); modified semi-implicit schemes evolve each Fourier mode exactly as an Ornstein–Uhlenbeck process (Lord and Tambue, 2016); splitting schemes preserve state-space structure in chemical Langevin dynamics (Jovanovski et al., 2025; Ulander, 2024). These are the honest competitors for stiff reaction–diffusion SPDEs. They remain pathwise; the per-mode-exact philosophy, however, is the same one that makes our decoupled M/V-block stepping work (§4).

2.3 Moment methods and closures

The moment hierarchy approach has a long history in kinetic theory and population dynamics; the closure problem is its central obstruction. Gaussian (Wick) closure is exact for linear dynamics driven by Gaussian noise and otherwise a weak-noise approximation that fails for bimodal or intermittent responses — a point developed carefully in the companion primer (moment_hierarchy_primer.tex)

and used as the scope boundary here. The NIDC framework’s position is deliberately narrow: it claims nothing about closures, only that *when a moment system closes, it can be solved to arbitrary order on cheap stencils*.

2.4 Multirate time stepping

Resolving competing optimal steps (§4) is a multirate problem. The classical theory of Gear–Wells (Gear and Wells, 1984) and Fok–Rosales (Fok and Rosales, 2010) gives the interpolation-order bookkeeping used in §4 (cubic Hermite suffices through $p = 2$); Richardson-style extrapolation (Oates et al., 2024; Debrabant and Lang, 2011) is the post-processing alternative that buys one order at the cost of two coordinated solves and shared-noise synchronization, whereas NIDC buys two orders inside a single solve. Recent multirate partitioned Runge–Kutta methods (Roberts et al., 2018) are the modern heavy machinery for the same problem class.

2.5 Positioning summary

Method	Moment accuracy	Cost to moments
Euler–Maruyama (Lord et al., 2014)	weak order 1	$O(M)$ paths, $O(M^{-1/2})$ error
Milstein (Leonhard and Rößler, 2014)	weak order ≥ 1	same, more work per path
Richardson on FD+EM (Oates et al., 2024)	weak order 2	two solves, shared noise
Exponential integrators (Lord and Tambue, 2010)	inherits time integrator	pathwise + MC
OTS-NIDC (this work)	4th order (moment PDE solve)	one deterministic solve

The comparison is deliberately asymmetric: the pathwise methods deliver information (paths) that OTS-NIDC does not attempt. The claim is narrower and cheaper: *if the deliverable is moments, and the noise model closes the hierarchy, then the moments are a deterministic PDE problem and should be solved like one*.

3 The method I: scalar OTS-NIDC on the reaction–diffusion family

Throughout, $\delta_x^2 u_j = (u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$ is the centered second difference and refinement means $\Delta x \rightarrow 0$ with Δt slaved to Δx .

3.1 The scalar family and the optimal time step

Consider the deterministic family

$$u_t = D u_{xx} + \alpha u, \quad D > 0, \alpha \in \mathbb{R}, \quad (3)$$

whose stochastic motivation (the moment equations of (1), with $\alpha = \frac{k(k-1)}{2}\sigma^2$) is deferred to §5. The pure diffusion case $\alpha = 0$ is exactly Chu’s setting (Chu, 2011).

Apply forward Euler in time and the three-point stencil in space. Taylor expansion of both sides, followed by repeated differentiation of (3) to convert time derivatives of u into space derivatives (the operators $D\partial_{xx}$ and α commute, giving the binomial identity $u_{t^k} = (D\partial_{xx} + \alpha)^k u$), yields the

local truncation error organized by derivative type:

$$\tau = \underbrace{\left(\frac{D^2\Delta t}{2} - \frac{D\Delta x^2}{12}\right)u_{xxxx}}_{\text{Group 1}} + \underbrace{\frac{\alpha D \Delta t}{2}u_{xx}}_{\text{Group 2}} + \underbrace{\frac{\alpha^2 \Delta t}{2}u}_{\text{Group 3}} + O(\Delta t^2, \Delta x^4). \quad (4)$$

Only Group 1 couples Δt to Δx . Setting it to zero:

Theorem 1 (Optimal time step (Chu, 2011)). *For (3) discretized by forward Euler / centered differences, the unique step size annihilating the leading (u_{xxxx}) truncation term is*

$$\Delta t_{\text{opt}} = \frac{\Delta x^2}{6D} \quad (5)$$

and no single step annihilates both the u_{xxxx} and u_{xxxxx} truncation terms, since the level-2 condition is $\Delta t = \Delta x^2/(\sqrt{60}D) \neq \Delta t_{\text{opt}}$.

(Proof: one-line cancellation in (4); uniqueness by the level-2 computation. Full algebra in `sign-change.tex` §1.)

Proposition 2 (α -independence). Δt_{opt} is independent of the reaction parameter α : the cancellation operates on the diffusive group alone. For the stochastic hierarchy this makes Δt_{opt} independent of the noise amplitude σ and of the moment order k .

Corollary 3 (One step governs the hierarchy). *All moment equations $M_{k,t} = D M_{k,xx} + \frac{k(k-1)}{2}\sigma^2 M_k$ of the scalar stochastic heat equation share the single optimal step $\Delta t_{\text{opt}} = \Delta x^2/6D$. Only the defect corrections differ, through explicit α_k -dependent coefficients.*

Status: [P](algebraic identity; numerically exercised by every scalar run below). The same mechanism is the level-1 shadow of Chu–Lambers’ variable-coefficient OTS (Chu and Lambers, 2009): α -dependence survives only where the coefficient of the *leading* spatial operator varies.

3.2 The first-level correction, and the sign that matters

At $\Delta t = \Delta t_{\text{opt}}$ the Group-1 term vanishes and the surviving residual is

$$\tau|_{\Delta t_{\text{opt}}} = \frac{\alpha \Delta x^2}{6} u_{xx} + \frac{\alpha^2 \Delta x^2}{12D} u + O(\Delta x^4). \quad (6)$$

Remark 4 (Sign convention). The truncation error τ as assembled in (4) is the *forward* residual: exact – scheme = $\Delta t \tau + \dots$. Cancelling it therefore requires *adding* $\Delta t \tau$; the earlier manuscript version subtracted it and thereby doubled the leading error. The corrected correction is

$$C_j^{(1)} = +\frac{\alpha \Delta x^4}{36D} \delta_x^2 u_j^n + \frac{\alpha^2 \Delta x^4}{72D^2} u_j^n \quad (7)$$

and the same flip applies to the general p -level tower (9). Independent confirmation: doc-sign converges at order 2.01, corrected sign at 4.01 (Table 1). Full algebra: `sign-change.tex`.

Theorem 5 (First-level accuracy). *The corrected scheme*

$$u_j^{n+1} = u_j^n + \Delta t_{\text{opt}} \left[D \delta_x^2 u_j^n + \alpha u_j^n \right] + C_j^{(1)} \quad (8)$$

has step error $O(\Delta x^6)$ and global order 4: the correction $C^{(1)}$ is itself $O(\Delta x^4)$, so its evaluation with the second-order operator δ_x^2 contributes only $O(\Delta x^6)$ to the step — the three-point stencil is mathematically sufficient at $p = 1$.

Table 1: Scalar $p = 1$ convergence, $u_t = Du_{xx} + \alpha u$ with $D = 0.1$, $\alpha\sigma^2 = 0.25$, Neumann BCs, $T = 0.05$; L^2 error against the exact cosine-series solution. Data: `code/implementation/results.md` (generated by `convergence_test.py`, 2026-07). All methods use the same grid and step size Δt_{opt} at each level.

Method	$N = 32$	$N = 64$	$N = 128$	$N = 256$
OTS-NIDC ($p = 1$)	8.54e-09	5.26e-10	3.27e-11	2.04e-12
Forward Euler	1.23e-05	3.02e-06	7.52e-07	1.88e-07
4th-order FD + FE	1.52e-05	3.79e-06	9.45e-07	2.36e-07
Reference expm	2.77e-05	6.81e-06	1.70e-06	4.23e-07
<i>Measured order (log-log slope, mean over levels):</i>				
OTS-NIDC	4.01	(FE / 4th-order FD / expm: 2.00–2.01)		

Status: [V]. Measured orders and errors, with the harness described in Appendix A:

The 4th-order-FD row carries the paper’s cleanest negative control: high-order spatial accuracy alone buys nothing when the step size is slaved to Δx^2 — the $O(\Delta t)$ time error caps the whole scheme at second order. The boost must come from the time-step structure; that is the method’s thesis, verified.

3.3 The p -level tower and the stencil bound

The general correction is generated by powers of the operator, with the sign of Remark 4:

$$C_j^{(\leq p)} = + \sum_{\ell=1}^p \frac{\Delta t_{\text{opt}}^\ell}{(\ell+1)!} (D\mathcal{D}_2 + \alpha)^\ell u_j^n, \quad (9)$$

where $\mathcal{D}_2$ is a spatial operator whose accuracy must be raised as ℓ grows. Spatial discretization error breaks the binomial symmetry of the time expansion, giving the stencil-scaling bound:

Proposition 6 (Stencil bound; sketch). *Global order $2p + 2$ requires corrections evaluated on at least a $(2p + 3)$ -point stencil: $p = 1$ needs the 3-point main stencil plus a 5-point evaluation inside $C^{(1)}$ when full fourth-order spatial accuracy is demanded of the correction itself; $p = 2$ needs a 7-point stencil to represent the uncanceled u_{xxxxxx} residual.*

Status: [O] for $p \geq 2$ numerics (no harness yet; the bound’s derivation is a consistency argument in OTS-NIDC.md §6). The $p = 1$ row is [V] via Table 1.

3.4 Stability

Proposition 7 (Von Neumann stability). *With $s = \sin^2(k\Delta x/2)$, $\mu = D\Delta t/\Delta x^2$, $\eta = \alpha\Delta x^2/D$, the amplification factors are $G_{\text{FE}} = 1 - 4\mu s + \mu\eta$ and*

$$G_{\text{NIDC}} = 1 - 4\mu s + \mu\eta - \frac{\eta}{9}s + \frac{\eta^2}{72}.$$

For the moment hierarchy $\eta \geq 0$ and the grid-scale stability bound tightens from $\mu \leq 2/(4 - \eta)$ to $\mu \leq (2 - \eta/9 + \eta^2/72)/(4 - \eta)$: a relative shrink of $1 - \eta/18 + O(\eta^2)$, i.e. $O(\Delta x^2)$, about -1.6% at the coarse $\eta = 0.3$ test point. The optimal step $\mu = 1/6$ sits a factor of 3 inside the CFL ceiling for both schemes.

Status: [V](analytic + numerical check in `code/stability/`; the earlier “expands the CFL” conclusion was an artifact of the pre-correction sign and has been fixed in the same release). The correction is stability-neutral to first order; it never threatens the optimal step. Note the honest asymmetry: this is a *scalar* analysis; the coupled discrete schemes of §4 have no independent stability proof (open item 2).

3.5 Efficiency profile

The method’s cost argument is architectural, not just arithmetic: versus multi-stage Runge–Kutta, OTS-NIDC is single-pass (one read, one write of the state vector; no stage storage, no stage barriers); versus compact schemes, it is strictly local (no tridiagonal solves), so it maps onto GPUs and bandwidth-bound hardware without global communication. At $N = 256$ the verified scalar runs show $\sim 10^3$ – $10^4 \times$ lower error than forward Euler at identical grid and step [V]— order wins over constants at refinement.

4 The method II: coupled reaction–diffusion systems

We now couple species — the deterministic core needed for the stochastic two-block system of §5, and the natural home of the paper’s hardest verification question.

4.1 Mean-field and second-moment blocks

For N species with diagonal diffusion, linear reaction coupling, and (here) zero noise — the deterministic skeleton of the moment systems to come —

$$\partial_t u^{(i)} = D_i \partial_{xx} u^{(i)} + \sum_j R_{ij} u^{(j)}. \quad (10)$$

Taking $u^{(i)} = \mathbb{E}[\cdot]$ of a stochastic system instead produces the *mean-field (M) block* of §5; the pairwise products produce the *second-moment (V) block* with diffusion coefficients $D_{ij} = D_i + D_j$. Both have the identical structure of (10) with different D ’s and source terms, so the analysis of this section applies to each block verbatim.

The per-equation LTE analysis of §3 goes through with $D \rightarrow D_i$ (M block) or $D \rightarrow D_{ij}$ (V block), the reaction coupling entering only through the analogues of Groups 2–3. The optimal steps are

$$\Delta t_{\text{opt}}^{(i)} = \frac{\Delta x^2}{6D_i} \quad (\text{M, species } i), \quad \Delta t_{\text{opt}}^{(ij)} = \frac{\Delta x^2}{6(D_i + D_j)} \quad (\text{V, pair } i, j), \quad (11)$$

with first-level corrections $C^{(1),(i)}$ and $C^{(1),(ij)}$ of the same algebraic form as (7), derived term-by-term in OTS-NIDC.md §8–9. In the moment hierarchy $D_i = D$ these become $\Delta x^2/6D$ and $\Delta x^2/12D$ — a strict **2:1 ratio**, independently verified in `notes/Di_eq_D_verification.md`: measured $\Delta t_M/\Delta t_V = 2.0000$ at $N = 32, 64, 128$ [V].

4.2 The M/V decoupling

Proposition 8 (Exact block decoupling). *For affine drift with linear multiplicative noise (the class of §5), the M-system is closed in $\{m_i\}$ and the V-system is closed in $\{V_{ij}\}$: neither block feeds the other, for any D_i, R_{ij}, σ_{ik} .*

Proof sketch. In the M-block, every stochastic contribution is $\mathbb{E}[dW_k] = 0$: no second-moment term can appear. In the V-block, Itô’s product rule contributes the quadratic covariation $\sum_k \sigma_{ik} \sigma_{jk} u_k^2 dt$, whose expectation is a sum of second moments V_{kk} — the drift being affine and the noise being linear, no third moments are generated. Closure is a structural consequence of (affine drift) $\times$ (linear noise); it is exactly what fails for square-root (chemical Langevin) noise, whose hierarchy closes at no finite order (Jovanovski et al., 2025). $\square$

Status: [P]. Consequence: the two blocks are two independent programs — no communication, no interpolation, no synchronization constraint *between* blocks, for any diffusivities. The residual problem is *within*-block: species i and j with $D_i \neq D_j$ want different Δt_{opt} ’s.

4.3 Competing optimal steps within a block

The two failure modes of a naive single step frame the problem.

Failure 1: shared step without correction. If species 1 (fast, D_1) is forced onto species 2’s optimum $\Delta t^* = \tau_2$, its Group-1 cancellation breaks. The one-step identity

$$e(\Delta t^*) = \Delta t^* D_1 \tilde{u}_{xxx} \left[\frac{\Delta t^* D_1}{2} - \frac{\Delta x^2}{12} \right] + O((\Delta t^*)^3) \quad (12)$$

is exact for every Δt^* , and writing $\Delta t^* = \tau_1 + \epsilon$ turns the bracket into $\epsilon D_1/2$ *identically* (no Taylor truncation in ϵ). Hence:

Theorem 9 (Exact mismatch correction). *For any shared step $\Delta t^* = \tau_1 + \epsilon > 0$ that is stable for species 1, subtracting (12) (with $\tilde{u}_{xxx}$ from a 5-point stencil) restores species 1’s full accuracy. The correction is algebraic, not asymptotic, in ϵ — including ϵ of the same size as τ_1 itself.*

Status: [V](worked example $D_1 = 3, D_2 = 1$ at $r = 3$, `examples/test_regime1_D1.3_D2.1.py`; correction there is grid-independent — the Δx^4 cancels, leaving pure stencil weights).

Failure 2: naive splitting. Lie–Trotter splitting with the coupling frozen over a macro step H incurs a sustained $O(H)$ forcing error; accumulated over T/H cycles this is a global $O(H) = O(\Delta x^2)$ error — worse than the uncorrected base scheme. Freezing is not viable; coupling data must be interpolated (order $m \geq p$ by the Gear–Wells bookkeeping (see Thm. 11’s setting), cubic Hermite sufficing through $p = 2$ (Gear and Wells, 1984; Fok and Rosales, 2010)).

Theorem 10 (Regime boundary). *Forcing the fast species onto the slow species’ optimal step $\Delta t^* = \tau_2 = \Delta x^2/6D_2$ is stable for the fast species iff*

$$r := D_1/D_2 \leq 3.$$

For $r > 3$ the shared step is unconditionally unstable and subcycling is mandatory.

Proof sketch. FTCS stability needs $\Delta t^* \leq \Delta x^2/2D_1$; substituting $\Delta t^* = \Delta x^2/6D_2$ gives $r \leq 3$. The threshold is grid-independent because both sides scale as Δx^2 . $\square$

Status: [S]— derived from plain FTCS; the corrected scheme’s stability region (Prop. 7 is scalar-only) has not been re-derived, so the boundary is supported, not proven, for the corrected coupled scheme (open item 2).

Theorem 11 (Remainder-step uniformity). *Under subcycling with macro step $H = \tau_2$, $n = \lfloor r \rfloor$ exact substeps of τ_1 , and one remainder substep $\rho = \tau_1 \text{frac}(r)$: the remainder’s leading error $e(\rho) = -\frac{D_1^2}{2} \rho(\tau_1 - \rho) \tilde{u}_{xxxx} + \dots$ vanishes at both $\text{frac}(r) \rightarrow 0$ and $\rightarrow 1$, and obeys the uniform bound $|e(\rho)| \leq \frac{D_1^2 \tau_1^2}{8} |\tilde{u}_{xxxx}| = O(\Delta x^4)$ for all $\text{frac}(r) \in [0, 1)$.*

Proof sketch. $\rho(\tau_1 - \rho)$ is a downward parabola on $[0, \tau_1]$, maximal at the midpoint, zero at both ends. No value of the step ratio is singular; the discriminator is how *often* a mismatched step is taken (once per cycle here), not how close r is to an integer. $\square$

Status: [P]for the leading (Group-1) term — explicitly a leading-only result; the Group-2/3 book-keeping at arbitrary ρ is open (item 3). Verified end-to-end at the integer ratio $r = 4$ ($\rho = 0$, four exact substeps) in `examples/test_regime2_D1_1.D2_4.py` [V]for the regime-2 algorithm as a whole.

4.4 The coupled contradiction: the ledger in action

Corollary 12 (Under separate stepping; **contradicted**). Claim as originally drafted: *evolving the M -block at $\Delta t_{\text{opt}}^{(i)}$ and the V -block at $\Delta t_{\text{opt}}^{(ij)}$ (2:1 in the hierarchy) preserves fourth-order accuracy in both blocks with three-point stencils.*

This claim is tagged [C]. The harness `coupled_verify.py` (`code/verify/21ratio/`) implements the §8.9/§9.8 corrections faithfully against a validated high-resolution reference and measures **order** ≈ 1.99 (**M**) and **2.02** (**V**) — second order, not fourth — while confirming (i) the 2:1 ratio exactly, and (ii) that the corrections reduce error constants by 3–40 $\times$ over plain forward Euler, so the corrections are directionally right but an $O(\Delta x^2)$ defect survives. Two hypotheses, not yet distinguished:

- (A) **Derivation gap:** the coupled corrections miss a cross-term (e.g. the discrete δ_x^2 acting on a *different* coupled field inside $C^{(1)}$ injects an $O(\Delta x^2)$ error that caps the scheme, unlike the scalar case where the correction acts on its own field), or the synchronization of clocks leaves a residual.
- (B) **Harness bug** in exactly the coupled-field plumbing.

We report both because the ledger does not allow burying either: until this is resolved, the coupled fourth-order claim is *withdrawn from* the paper’s proven core, and the honest statement is: *scalar fourth order is verified; coupled fourth order is open, with a reproducible contradiction and a candidate mechanism*. The discrepancy gates submission (§7).

5 The method III: stochastic reaction–diffusion via the moment hierarchy

Everything so far is deterministic. The stochastic payload enters through a single structural fact: *for the right noise class, the SPDE’s moments obey closed deterministic reaction–diffusion equations* — the exact family (3) – (10) of §3–4.

5.1 The hierarchy and its exact closure

Proposition 13 (Moment closure for linear multiplicative noise). *Let u solve (1) (Itô, space-time white noise, linear multiplicative noise $\sigma u \dot{W}$). Then $M_k = \mathbb{E}[u^k]$ satisfies the closed deterministic reaction–diffusion equation*

$$\partial_t M_k = D \partial_{xx} M_k + \underbrace{\frac{k(k-1)}{2} \sigma^2}_{=\alpha_k} M_k, \quad k = 1, 2, \dots \quad (13)$$

i.e. (3) with $\alpha = \alpha_k$. More generally, for the N -species linear system $du_i = (D_i \partial_{xx} u_i + \sum_j R_{ij} u_j) dt + \sum_k \sigma_{ik} u_k dW_k$, the mean fields $m_i = \mathbb{E}[u_i]$ obey the closed M -system (10) and the covariances $V_{ij} = \mathbb{E}[u_i u_j]$ obey the closed V -system

$$\partial_t V_{ij} = (D_i + D_j) \partial_{xx} V_{ij} + \sum_k R_{ik} V_{kj} + \sum_k R_{jk} V_{ik} + \sum_k \sigma_{ik} \sigma_{jk} V_{kk}. \quad (14)$$

Proof sketch. Apply Itô’s formula (finite-dimensional, then per grid cell / Fourier mode) to $\Phi(u) = u^k$ and to the product $\Phi(u) = u_i u_j$. The drift term survives expectation; the Itô integral against dW has zero mean; the quadratic variation contributes $\frac{1}{2} \Phi''(u) \sigma^2 u^2 dt$, which for $\Phi = u^k$ is $\frac{k(k-1)}{2} \sigma^2 u^k dt$ — proportional to M_k itself, which is the closure. For (14) the product rule adds the covariation $\sum_k \sigma_{ik} \sigma_{jk} u_k^2 dt$, whose expectation is the V_{kk} term. Full derivations: `OTS-NIDC.md` §7–9; `moment_hierarchy_primer.tex`. $\square$

Status: [P]. Two scope qualifiers carry the proposition’s honest weight. (i) *Discretized regime:* for genuine space-time white noise the continuum second moment diverges ($\delta(0)$); the closure is well-posed on the spatial discretization or for trace-class (colored) noise, which is the regime every finite-difference method inhabits anyway. (ii) *Closure is structural:* exact for additive or linear-multiplicative noise with affine drift; fails for polynomial reactions of degree ≥ 2 (Fisher–KPP: m_1 needs m_2 , m_2 needs m_3 , ...) and for nonlinear noise (chemical Langevin closes at no finite order (Jovanovski et al., 2025)).

5.2 What the deterministic theory immediately buys

Proposition 13 makes §3 apply verbatim to every moment equation. The stochastic highlights:

1. **One step for the whole hierarchy** (Cor. 3): $\Delta t_{\text{opt}} = \Delta x^2 / 6D$ governs every M_k simultaneously; σ and k appear only in the correction coefficients (7) — e.g. the mean ($\alpha = \frac{1}{2} \sigma^2$) and second moment ($\alpha = \sigma^2$) are advanced by the same step, different corrections.
2. **The Itô drift is a defect, not an obstacle:** the noise-induced reaction term is explicitly computable, so it slots into $C^{(1)}$ exactly as any reaction term would. This is the conceptual bridge of the paper: Itô calculus converts noise into reaction, and OTS-NIDC’s cancellation is indifferent to where the reaction came from.
3. **Two blocks, two steps, no coupling** (Prop. 8): mean and second-moment systems decouple exactly; in the hierarchy $D_i = D$ the 2:1 step ratio is exact [V].
4. **Weak, not strong, by construction:** the method never samples $\dot{W}$, so the Davie–Gaines strong-order-1/2 ceiling (Davie and Gaines, 2001) governs a problem we do not solve. The delivered object is fourth-order convergence of the computed moment fields to the true moments of the closed system [V] (scalar, Table 1); conversion to statements about the SPDE law rides entirely on Proposition 13’s exact closure.

5. **Stability transfers:** the growing-moment case is exactly the $\eta \geq 0$ regime of Prop. 7 — the correction marginally *shrinks* the CFL window and is comfortably stable at Δt_{opt} [V].

5.3 The competitor, stated fairly

Euler–Maruyama on (1) estimates M_k from M paths with error $O(\Delta t) + O(M^{-1/2})$ per point; our implementation (`baselines.py`, `euler_maruyama_moment`) exists for exactly this comparison [S]¹. The structural comparison, however, is already decisive and does not need the figure: for the *moments alone*, the pathwise route pays M path-solves plus Monte Carlo variance to approximate what the moment route computes deterministically — and the moment route’s solver is fourth-order where EM is weak-first. The pathwise route is irreplaceable when paths are the product (extremes, hitting, filtering); for moment delivery under exact closure it is dominated.

5.4 Where the noise model ends

Three extensions are visible from inside the current theory, all open:

- **Colored noise** (trace-class Q): the closure proof goes through with $\sigma^2 \rightarrow$ the covariance trace structure; the M/V systems acquire nonlocal reaction kernels. The OTS cancellation, being diffusive-local, should survive with kernel-aware corrections [O].
- **Noise discretization interplay:** how Wiener-increment sampling on the grid (piecewise-constant vs. linear-in-time noise, (Lord and Petersson, 2022)) interacts with the NIDC cancellation — for the moment route there is no sampled noise, which is precisely the advantage; the question becomes how to *validate* moment solvers against pathwise references fairly [O].
- **Boundary conditions for V :** Neumann data on u induce $\partial_n V_{ij} = 0$ on $\partial\Omega \times \Omega \cup \Omega \times \partial\Omega$; used but not proven in the coupled harness [O].

6 The nonlinear frontier: Cole–Hopf, stochastic Burgers, and KPZ

Everything above lives inside the closure-exact class: linear reaction, linear (or additive) multiplicative noise. This section maps the road out — and is deliberately the part of the paper where the ledger does the most work, because the beautiful connection here is also the one where the current framework *cannot follow*.

6.1 Stochastic Burgers is a stochastic heat equation in disguise

The forced stochastic Burgers equation (in 1D, written for the slope u) is

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u \quad \Longleftrightarrow \quad u = -2\nu \partial_x \log \phi. \quad (15)$$

Under Cole–Hopf, ϕ solves a stochastic heat equation with *multiplicative* noise. In its simplest well-posed version (Itô, with the noise entering as $\sigma \phi \xi$ plus the Itô correction absorbed into the equation for ϕ), ϕ is *exactly* the linear-multiplicative-noise model (1) — the protagonist of §5:

¹The EM baseline is implemented and its noise scaling ($\sigma u \sqrt{\Delta t / \Delta x} Z$ per cell per step) is the standard space-time white noise discretization; the head-to-head convergence figure against it is not yet in the results ledger — it is scheduled with the benchmarking gate of §7, per the project’s own decision to hold benchmarking until scoping stabilized.

Theorem 14 (Cole–Hopf transfer of weak schemes). *Let ϕ solve the linear multiplicative-noise stochastic heat equation, so its moments $M_k^\phi = \mathbb{E}[\phi^k]$ satisfy the closed family (13) and are computable by OTS-NIDC to fourth order (Table 1). Then every moment of the exponential functional $\phi^k(x, t)$ of the Burgers velocity field — in particular the k -th moments of $u(x, t) = -2\nu \partial_x \log \phi(x, t)$ that are expressible as derivatives of $\log M_k^\phi$ or of ratios of ϕ -moments — inherits that accuracy, because u is a fixed smooth functional of the moments and the scheme converges in exactly the norm that functional reads.*

Proof sketch. Mild solutions: $\phi(t) = S_t \phi_0 \cdot \mathcal{E}_t$ with $\mathcal{E}_t = \exp(\sigma W_t - \frac{1}{2} \sigma^2 t \cdot)$ the stochastic exponential (a strictly positive martingale field); hence $\phi > 0$ and $\log \phi$ is smooth wherever ϕ is. Moments of ϕ are the closed hierarchy; derivatives of log-moments are smooth functions of those moments, so a solver converging at order q in the moments yields order- q accuracy in any fixed derivative of their logarithms. The Cole–Hopf identity converts those to moments of Burgers observables. $\square$

Status: [P]. Why this matters: it moves *two* canonical nonlinear objects — Burgers velocity statistics and log-normal-type fields — inside the reach of a linear theory, without any moment truncation. Note the honest fine print: $\mathbb{E}[u]$ and higher u -moments involve $\mathbb{E}[\partial_x \phi / \phi]$, a *ratio* of moments, not a moment — the transfer in Theorem 14 is clean for quantities that are smooth functionals of the M_k^ϕ (e.g. $\partial_x \log \mathbb{E}[\phi^k]$, structure functions built from derivatives of log-moments), and requires ratio-variance care otherwise. Making this precise and tested is open (item 6).

6.2 The KPZ window: what stays finite when the moments leave

The Kardar–Parisi–Zhang equation is the log of the same field: $u = \log \phi$ turns the multiplicative-noise heat equation into

$$\partial_t u = \nu \partial_{xx} u + \frac{\lambda}{2} (\partial_x u)^2 + \eta, \quad (16)$$

with η effective white noise and the Itô drift folded into the nonlinearity. Here the relationship between finiteness and observability *inverts*, and this inversion is the section’s central observation.

Proposition 15 (KPZ window). *In 1D, KPZ itself is well-posed (the Cole–Hopf solution is the definition of its solution), and its natural observables — the height field u , its roughness $\text{Var}[u]$, spatial correlations of $\log \phi$ — remain finite and well-defined for all time. It is the raw moments of ϕ , $M_k^\phi = \mathbb{E}[\phi^k]$, that diverge (log-normal tails, $M_k^\phi \sim e^{\frac{1}{2} k^2 \sigma^2 t}$): beyond a time of order $t_* \sim \log K / \sigma^2$ even the K -th moment overflows any fixed precision. The divergence is a property of the observable M_k^ϕ , not of the dynamics.*

Proof sketch. ϕ is the stochastic exponential: $\phi = \exp(\sigma W_t - \frac{1}{2} \sigma^2 t)$ applied to the diffused initial data, so $\log \phi$ is Gaussian at each point with variance $\sigma^2 t$ (per mode, with diffusive decorrelation). $\mathbb{E}[\phi^k]$ then carries the $\exp(\frac{1}{2} k^2 \sigma^2 t)$ log-normal factor — finite for each fixed t, k , but overflowing IEEE double precision ($\sim e^{709}$) at $t \approx 1418 / (k^2 \sigma^2)$, i.e. at practical times for even moderate $k\sigma$. Meanwhile $u = \log \phi$ has bounded Gaussian moments for all t . $\square$

Status: [S] (literature + representation argument; no numerics in this repo). The practical reading for this paper:

- **Work in u -space observables.** Roughness, correlation functions of u , and log-moment derivatives stay in the solver’s finite window indefinitely; the recommended solver target for KPZ-adjacent problems is the log-moment hierarchy, not the raw hierarchy.

- **But the closure is gone.** The equation (16) for u is *nonlinear in the field*, so the moment hierarchy of u does not close: $\mathbb{E}[(\partial_x u)^2]$ is a second moment feeding the mean equation, and the tower reopens. The exact-closure theorem (Prop. 13) does not reach KPZ; Gaussian closure fails precisely here (log-normal intermittency is the textbook failure mode). Nothing in §3 applies to (16) directly.
- **The window is nonetheless real.** What the theory *can* deliver inside the window: (i) ϕ -moments below the overflow time — K moments to time $t \sim \log K$, so a tower of solves extends the window logarithmically; (ii) all u -space statistics of the *linear* problem (no KPZ nonlinearity), exactly; (iii) the Cole–Hopf bridge (Thm. 14) for the observables it covers. What it cannot yet deliver: direct high-order weak integration of the KPZ nonlinearity (16) itself [O]— the honest frontier.

6.3 Why this section is in the paper

A living paper should show its edges. The Cole–Hopf/KPZ connection is the most-cited potential application of moment-level solvers (turbulence analogues, directed polymers, growth interfaces), and it is also the setting where the paper’s core machinery is *not yet* valid beyond the linear window. Rather than claim the frontier or quietly omit it, we state the transfer theorem that does hold ([P]), the window proposition that scopes it ([S]), and the open problem that gates the rest (item 7): *a closure scheme for the $(\partial_x u)^2$ moment tower that is exact in the same sense Prop. 13 is exact for linear noise — or a proof that none exists.*

7 Open problems and the submission gate

7.1 Open items, ordered by what they block

1. **[C]resolution (blocks submission).** Diagnose the coupled second-order measurement (Cor. 12): distinguish mechanism (A) — the coupled $C^{(1)}$ evaluates δ_x^2 on a *different* field, injecting an uncorrected $O(\Delta x^2)$ term — from mechanism (B), a harness bug, by re-deriving the coupled correction on the discrete level (the scalar proof’s step where $\delta_x^2 u \rightarrow u_{xx} + O(\Delta x^2)$ is used *inside* the correction requires the mixed-field analogue checked term by term). Either outcome advances the paper: (A) yields a corrected coupled scheme; (B) restores Cor. 12.
2. **Coupled stability proof.** Von Neumann for the subcycled + interpolated + corrected scheme, and re-derivation of the $r^* = 3$ boundary from the corrected scheme’s own stability region (Prop. 7 covers the scalar case only).
3. **Higher-order remainder bookkeeping.** Group-2/3 residuals for steps of arbitrary size ρ (the uniform bound of Thm. 11 is leading-term only); floating-point conditioning of wide-stencil mismatch corrections (Δx^{-2} -scaled cancellations).
4. **$p \geq 2$ harness.** Implement the 6th-order ($p = 2$, 7-point) tower; confirm the stencil bound of Prop. 6 numerically; measure where wide stencils stop paying (vs. grid refinement cost).
5. **Weak-order statement and MLMC coupling.** A clean theorem statement: OTS-NIDC delivers order $2p + 2$ in Δx (hence order $p + 1$ in Δt) for the closed moment system — a weak-order-2 base case at $p = 1$. Coupled with multilevel Monte Carlo, a weak-order-2 base eliminates the $\log \epsilon$ factor against EM’s $O(\epsilon^{-2} \log^2 \epsilon)$ complexity [O].

6. **Cole–Hopf transfer, made computational.** Turn Thm. 14 into a numerical experiment: solve the closed ϕ -moment hierarchy, form $\partial_x \log M_1^\phi$ and higher log-moment observables, validate against direct Monte Carlo Burgers simulations; quantify the ratio-variance fine print.
7. **KPZ closure (the frontier).** An exact (or provably faithful) treatment of the $(\partial_x u)^2$ tower — or a no-go result mirroring the Fisher–KPP obstruction. Candidate directions: Wick-ordered KPZ in 1D (where the renormalized equation is well-posed and the nonlinearity is a martingale perturbation), or working entirely with the log-moment hierarchy inside the Prop. 15 window.
8. **Gray–Scott and the coupled benchmark pin.** The project’s own scoping decision holds benchmarking until the focus question settles; the leading candidate benchmark is a coupled stochastic RD system (Gray–Scott-class) exercising the $D_i \neq D_j$ machinery of §4 end to end.

7.2 The submission gate

This paper is *almost* ready for submission, by its own ledger. The gate is explicit and checkable:

- **Gate 1 (hard):** resolve item 1 — no submission with an open [C] on a headline claim.
- **Gate 2 (hard):** item 2 for whatever coupled scheme survives Gate 1.
- **Gate 3 (soft, scope-setting):** items 4–5 may ship as future work, but at least the scalar EM head-to-head (the footnote of §5) should be converted from [S] to [V].
- **Not gates:** 6, 7, 8 — these are the *next* paper’s core, included here as the frame it will sit in.

7.3 Reproducibility

Every [V]number in this paper is regenerated by scripts committed in the repository; Appendix A lists the exact commands. The ledger and this paper evolve together: a claim’s tag may only change together with a change-log entry naming the artifact that moved it.

A Reproducibility manifest

All artifacts live in the OTS–NIDC repository. Verified numbers were produced by the scripts below; each ledger entry in the front matter names its artifact.

Artifact	Command (under <code>code/</code>)	Ledger claims supported
<code>implementation/convergence_test.py</code>	<code>python convergence_test.py</code> (venv w/ numpy, scipy, matplotlib)	Scalar 4.01/2.01 orders (Table 1); FD-cap negative control
<code>implementation/baselines.py</code>	(imported by above; <code>euler_maruyama_moment</code> for EM)	EM baseline (S); noise-scaling convention
<code>verify/21ratio/coupled_verify.py</code>	<code>python coupled_verify.py</code>	2:1 ratio [V]; coupled order ≈ 2 [C]
<code>stability/stability_check.py</code>	<code>python stability_check.py</code>	CFL shrink factor; stability at Δt_{opt}
<code>examples/test_regime1_D1.3_D2.1.py</code> <code>test_regime2_D1.1_D2.4.py</code>	<code>python run_tests.py</code>	Regime-1 mismatch correction; regime-2 subcycling ($r = 4$)

Environments. Scalar and coupled harnesses ran under per-directory virtual environments (numpy/scipy). The stability analysis is analytic with a numeric spot-check ($\eta = 0.3$). Dates of the archived outputs are recorded in the respective `results.md` / verification files; the paper cites those archived outputs, and regeneration is expected to reproduce them to machine precision (deterministic seeds: `seed=12345` in `baselines.py`).

Honesty notes carried from the harnesses. (i) The coupled verification reference was itself validated against the known scalar exact solution before use. (ii) The regime-2 example uses an exact integer ratio ($r = 4$, $\rho = 0$); fractional ρ remains covered by the leading-order bound only (Thm. 11). (iii) Page numbers in the literature-pointer notes are approximate (form-feed counts); verbatim excerpts are preserved in `notes/Di_eq_D_literature_pointers.md`.

Compilation.

```
cd paper && latexmk -pdf main.tex    # auxiliary output under ../aux/
```

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